

Responsible Software - CheatSheet

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INTRO. Responsible Software. Software engineered in a responsible (ethical) way.
1. With a goal to do good. 2. While preventing avoidable negative impacts.
3. Taking people, other systems, social structures and our planet into account.
Active Responsibility. anticipating/preventing negative impacts before occurring.
Ethics. Principles/standards that guide human behavior in various contexts.
Metaethics. "what does it mean for an action to be right?"
Normative. "how should people act in general (moral principles)?"
Applied. (Our approach) "how can i act ethically in a real-world situation?"

Technical decisions can have *ethical* impli-**Ethical Decision Making.**
cations.
1. **Recognize**(knowledge+experience+habits)
Ethical Issue. any situation that may com-2. **Evaluate**(cognitive+reason+ffective) 3. **Act.**
promise respect of ethical value/principle. **Ethical Lenses.**
Ethical Agency. capacity to take actions that1. **Desirability.** product corresponds to what people want.
align with ethical values and principles. 2. **Viability.** product generates profits for a business.
Ethical Sensitivity. ability to 3. **Feasibility.** product can be created with existing or new
- recognize ethical issues (awareness) technology.
- identify their impact: who, what **Fourth criterion (Isaac et al.).** add **Ethicality.**

Ethical Dilemmas. decision between multiple alternatives, all conflict with ethical values/principles. *None of the options override the other, wrong-wrong choice.*

Moral side. **Cognitive side (Normative Ethics).**
Emotions should be used explicite-**Utilitarianism(Consequences).** choose the greatest good for greatest
ly/intentionally for responsible de- number
sign (Roesser). In response to moral
issues or motivating immoral be-
havior. **Deontology(Action).** do right according to universal principles
Virtue(Person). act as a virtuous person
Care(Relationships). mutual responsibility/care for each other
choice: do nothing – runaway trolley kill people, or intervening →
fewer people die. tests which moral principles justify action vs. inaction.

Ethical Decision Frameworks. **Example questions for brainstorming**
1. Identify the ethical issues -Who will use your product on a daily basis?
2. Get the facts -Who is in geographical proximity with the users of your product?
3. Evaluate alternative actions with **several ethical theories** -Who is in social proximity with the users of your product?
4. Choose an option and "test" it -Who is involved in the maintenance of your product?
5. Implement your decision and reflect on the outcome. -Who provides services that are needed to run your product?
-Who provides funding?
-Which communities are active in the application domain?

Stakeholder Individuals/groups/organizations/institutions/societies that might reasonably be affected by technology. **Direct Stakeholder.** *in contact with the product*

Indirect Stakeholder. affected through *consequences of the usage* of the software
Stakeholder analysis. Identifying every entity that can be affected by software
positively/negatively. 1. Find all stakeholders possible using questions.
2. Use Direct vs Indirect Chart to find new stakeholders.

Additional Methodologies: *Ethix.* add *unintended stakeholders*, *Mitchell.* add
attributes to stakeholders, *SAS.* use *different chart with scale for impact from product and influence on the product.*



Ethical Speculation. How to think, not what to think - anticipate potential impacts on stakeholders.
Create a fictional story in two parts. (*done at the early design phase*)

I. Dark and Pessimistic 1. Imagine a character that would be negatively impacted by the software 2. Write a pitch that summarizes their story. 3. Find a title.	II. Positive outcome 1. Identify 1 or 2 main ethical issues represented in your story. 2. Describe the immediate and future consequences. 3. Imagine a happy ending for character.
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SAFETY II. Levels of analysis. Macro (societies), Meso (groups/communities), Micro (individuals)
Internet Traffic. 99% submarine cables: 1.4M km, 4.7B users, 1.2 ZettaBytes (1.2 trillion GB).
Internet Benefits (+). Information access/dissemination, Communication, Social relations/organization, Identity/psychological development, Learning/cognitive development, Production/commerce.
Tension. Same features enable both benefits/harms → *social connection enables social harm, information access causes overload, identity spaces blur privacy boundaries.*
From micro to macro.
Types of Impacts: lots of individuals, Social structures/institutions, Different for different groups.
Areas: Markets/economy, Knowledge/worldviews, health/well-being, Politics/citizenship, Environment.
△controversy Complex measurements, causality hard to establish.

Disinformation. false info, with intent to deceive. Misinformation. false info. without intent to deceive Malinformation. true info. used with harmful intent. <i>fake news</i> – false information, with or without intent. Cognitive effects. how we think. Socio-affective effects. interaction with people and emotions. System 1. fast/emotional/ activated first as quick response. System 2. slow/logical/conscious analysis. Illusory truth effect. repetition increases perceived truthfulness of claims by enhancing familiarity and intuitive/shortcut-based thinking. Continued influence effect: misinformation continues to influence beliefs even after correction is received and accepted as true. Increased use of digital media shows greater exposure to misinformation, increased political polarization, decreased trust, and increased hate. <i>empirical results remain debated and sometimes exaggerated</i>	Global risks. impact on electoral processes (<i>Brexit, U.S. elections</i>), growth of distrust (government/media sources), risks of repression. <i>socio-technical phenomenon</i> Socio-affective factors - Source cues. perceived credibility (powerful/attractive sources more likely to be believed). - False consensus. overestimating how many people agree (<i>climate change denial.</i> ~ 6%) <i>in reality, perceived</i> ~ 50%). - Emotions. strong emotions content/receiver emotional state increases susceptibility. - Identity resonance. alignment with personal identity/values increases acceptance.
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Publishing. publicly available content, *financial/political/coordinated* strategic incentives.
Human publishers. Influential figures/ordinary users publishing for visibility/revenue, often using *click-bait*/participating in organized campaigns (*coordinated troll operations*).
Social bots. Automated accounts publishing at scale (1–15%), they generate *disproportionately large share* of misinformation.
Propagation. content spreading through *reposting/liking/algorithmic amplification*.
Human propagation. content reshared by real/fake accounts. engagement inflated by paid workers, while moderation reduces resharing (*account bans lowering repost rates*).
Genuine resharing factors. Resharing driven by *inattention, emotional reactions, and socio-affective motives* such as signaling group membership or provoking reactions (*outrage*).
Social bots in propagation. Automated accounts rapidly *repost* and *target influential or opposing users* to maximize visibility and conflict.
Recommender systems. *select, rank, display* content, optimize for *engagement (popularity bias)*.

Ethics Canvas. Structure stakeholder, impact, and **Edge Cases.** Foresee consequences/identify opportunities for unforeseen analyzing extreme scenarios.
When to use? Early software development *Brain-When to Use: At all stages,design/major updates/user storm, visualize, mitigate ethical impacts*
1. Identify Stakeholders: Who is affected?
2. Ethical Impacts: What are the impacts on them?
3. Actions: How to address these impacts?
Causal loop diagram. to explain complex system/s/improve performance/solve issues
focuses on specific issues (*amount of fake news, investment in an auto-detection system*)
Causal links are labeled with a + / – longer delays are a whole to reach a state/purpose; modeling simplifies represented with thicker arrows.

SAFETY I.
Security. Prevent the environment from affecting negatively the system. *△threat | Vulnerability scanning, ethical hacking, security audit, code review*
Safety. Prevent the system from affecting negatively the environment. *△hazard | Verification and testing, failure modes analysis, Redundancy*
Common challenge - human factor → consider software as **socio-technical system**
Internet, Autonomous vehicles, social media, human behavior and technical systems constanly influence each other.
Malfunction. software doesn't function/perform normally/as specified.
- Due to unintentional defect/fault/bug→critical consequences(△Life/Mission-critical systems)
Misuse/Abuse. attack, diversion, use of software for *nefarious purposes*.
Bad/threat actors. *malicious intent, authorized/unauthorized users*→direct/indirect impact.
Unintended use. use of software in a way not originally intended/not (officially) supported.
⊕ creates opportunities for innovation *use camera to scan documents* ⊖ *use llm to obtain factual information*
Intended use software used as intended. difficult to anticipate impact due to social △scale/time

User-Generated Content (UGC). any content posted by users online *images/videos/texts*
- Published publicly and (usually) not produced by professional/institution *wikipedia*
△ Toxic content. material harmful/offensive can have negative impact on individuals/communities
Types of impacts? Physical/Financial/Emotional/Relational.
Hate and Harassment.
Violence threats, Identity misrepresentation
Self-inflicted harm. Eating disorder promotion
Ideological harm.
Extremism/terrorism, Misinformation
Exploitation. Sexual abuse, Scams
Hate speech
- Discriminatory/pejorative of individual/group
- calls out real/perceived "identity factors" of individual/group
- conveyed through any form of (images/memes/...) *UGC platforms provide channels for hate speech. Easily produced, shared, anonymous, global/diverse audience, realtime, amplified by content recommendation algorithms*

Predicted	Actual	
	Positive	Negative
Positive	True Positive (TP)	False Positive (FP)
Negative	False Negative (FN)	True Negative (TN)

TPModel predicts positive and is correct
Accuracy = $\frac{TP+TN}{TP+TN+FP+FN}$
Positive Predictive Value (PPV) = $\frac{TP}{TP+FP}$
Negative Predictive Value (NPV) = $\frac{TN}{TN+FN}$
False Discovery Rate (FDR) = $1 - PPV = \frac{FP}{TP+FP}$
False Omission Rate (FOR) = $1 - NPV = \frac{FN}{TN+FN}$
False Positive Rate (FPR) = $\frac{FP}{FP+TN}$
False Negative Rate (FNR) = $\frac{FN}{FN+TP}$

Bad Actors Modelling Minimize vulnerabilities via risk assessment and mitigation.
When to Use: Design phase, prior to updates, audits, incident responses, and improvements.
What is a Bad Actor? A person/group exploiting a system for personal goals.
Motivations (might overlap):
- Money: Data theft, ransomware, sabotage by competitors.
- Politics: Misinformation/strategic sabotage/infrastructure attacks.
- Entertainment: Attacks on random systems, misinfo. spreading.
- Self-interest: Harassing messages, communication monitoring, location tracking.
- Ideas: Discrediting groups, data leaks, website defacement.
Steps: 1. Identify possible malicious users and their motivations.
2. Assess potential consequences for the system and users.
3. Develop measures to prevent these negative outcomes.

Harm Modelling anticipate/address potential harms users/stakeholders may encounter.
When to Use: At every stage of project, for every feature, until harm potentials are satisfactory.
Evaluate: 1. The category of use. 2. the types of harms. 3. the magnitude of harm
Categories of Use:
Harm from...
Intended Use: correct usage.
Unintended Use: unexpected usage.
Misuse/Abuse: malicious usage.
age.
Malfunction: system failures. ment.
Types of Harm:
Humans: Physical, emotional injury.
Allocation of Resources: Opportunity, economic losses.
Human Rights: Dignity, liberty, privacy losses; environmental impact.
Social Systems: Manipulation, social detriment.

FAIRNESS I. Data in Software. people described by **attributes** (stored in databases/files), can be *sensitive*.
⇒ *ethical implications* such as *confidentiality, privacy, and fairness.* data is an *imperfect representation of real world*, must be *simplified* to fit context.
Data protection law (GDPR) - Personal data: reveals info about *identity*, directly/indirectly.
- **Special category data:** banned by default. *racial/ethnic origin, pol. opinions, rel. beliefs, health data.*
Nondiscrimination Law forbids direct/indirect discrimination, when **apparently neutral** practice puts *specific groups* at disadvantage. (*Protected Group/Class: Category qualifying for special protection by law/policy*)
Proxy Attributes. Variables that **correlate** w/ sensitive attributes.
Risks & Outcomes. can lead to **indirect discrimination** (*harder to detect*).
- Software has discriminatory outcomes even in absence of sensitive attributes in data. (*high school quality*) - correlates w/ income/ethnicity → becomes proxy.
Fairness. Equality *opportunities*; operationalized by treating people same. Scope matters: groups not protected law still face ethical harm.

Unfairness. Discrimination, stereotyping, decision errors based irrelevant factors; causing negative impacts via biased software/processes (*women get lower credit limits despite higher income*).
Bias. Systematic deviation from standard. context-dependent, not always harmful, possible ethical issues.
affects Problem definition/ideas/solutions/planning; shaped by external factors (info/org. culture)
1. Pre-existing bias. Bias already present world, captured in data (*systemic bias*).
2. Representation bias. Data inaccurately reflects target population/subgroup (*Sampling: over/under-representation. Population: sampled pop. differs actual users*).
3. Measurement bias. Methods yield inaccurate values, oversimplify reality, differ accuracy across groups (*blood-oxygen devices more accurate white people vs darker skin due melanin*).
4. Cognitive bias. Human (often implicit) deviations rational judgment; affects info processing/interpretation; enters via users/designers.
Stereotypes. Generalized beliefs about group (*implicit/explicit; pos/neg/neutral*); ignore individual differences, reinforced by design assumptions.
Group effects. groups with same opinions/preferences → reinforce confirmation bias; (*devil's advocate strategy*) doesn't work → groups with heterogeneous preferences more balanced
→ group-think: phenomena where alternatives are avoided, pseudo-consensus
Sunk-cost fallacy. continuing course of action just due to past investment → prevents alternatives→product with design flaws → **Technical debt.** future costs to fix, **Ethical debt.** ethical consequences/harms

Inclusive Design. design software that benefits the widest range of users by addressing diverse abilities, languages, cultures, and differences.
When to Use: Early in soft. development phases.
1. Identify needed capabilities for use.
2. Identify and label "Non-Average Users" (NAUs).
3. Add unaddressed capabilities/new user categories.
4. Develop and report concrete inclusive solutions.
Aim: Ensure artifacts match the needs of diverse individuals beyond average users.
Capabilities. perception(vision,hearing,...), thinking, action(dexterity,...)
Analyzing Values. Identify/analyze/evaluate values embedded in technology and resolve value-based tensions. *What are Values?* principles/desirable goals for design, based on Schwartz's value theory.
1. Identify values/stakeholder values/artifacts values.
2. Analyze value-based benefits (supporting stakeholder values)/harms (opposing stakeholder values).
3. Map value tensions:
* Within the same value (e.g., benefit vs. harm).
* Between different values (interpret context to resolve).
* Tools: Use value-based tension maps and questionnaires to assess impacts.

FAIRNESS II. ML. techniques inferring patterns from data, generalizable to unseen data. Process. Iterate to find parameters of function that best represent pattern in data. Result. Function + function parameters.	
Training. Run learning algorithm to find parameters → Inference. compute results on previously unseen data. Regression. Output continuous values (<i>house price from size</i>). Classification. Output discrete values (<i>wine vs beer from alcohol level, color</i>). <i>Supervised.</i> Output labels given. (<i>labels created by humans; picture descriptors</i>) <i>Unsupervised.</i> No output labels (<i>clustering, dimensionality reduction</i>). Data dependence. ML depends on data → sensitive to biases; in supervised, labels human-made → not all issues attributable to ML alone. Pipeline. Problem definition → data collect + preparation → model dev + model eval + model deploy. Model dev. Choose model type, objective function, hyper-parameter tuning for performance → unfairness risk when optimizing accuracy only. Aggregation bias. Combine groups with different patterns → model performs poorly for some groups, or mainly for majority. Evaluation bias. Benchmark not representative of served population → (<i>Gender Shades: intersectional bias across gender and race</i>). Deployment bias. Mismatch between intended problem and actual use → socio-technical bias. Fairness metrics. Mathematical measures of fairness notions (<i>criticised for oversimplification; focus outcome</i>). Error parity. Error rates independent of group. Contains: 1. Error rate balance. Equality of error rates compared to actual values→FPR, FNR across groups. 2. Conditional use accuracy equality. Equality of predictive values compared to predicted values→PPV,NPV across groups. 3. Equal accuracy. Equality of accuracy across groups. Demographic parity(Equal acceptance rate). Positive prediction rate independent of group (<i>focus outcome</i>): TP + FP Total positive prediction rate (group A) positive prediction rate (group B) should be 1. Impossibility result. Cannot satisfy all fairness metrics simultaneously. Group fairness criteria (GFC). Quantify equality between groups; bundle people treated differently → aim for group-level equality. → Mathematical impossibilities among parity measures, incl. with demographic parity. → Unfair world + imperfect classifiers → not all group fairness criteria hold simultaneously → choose priority notion of fairness. Individual fairness. Similar individuals should receive similar decisions; requires similarity metric, tricky. Causal analysis of fairness. Focus causes of unfairness and causal pathways from sensitive attributes to outcomes → find proxies via causal graphs. Improve fairness. Either modify data or modify model. Modifying the data. (<i>pre-processing</i>) If data biased, learned patterns biased. To reduce bias, <i>Fairness through unawareness</i> removes sensitive attributes from dataset, but reduces accuracy. If proxies exist, bias may persist. Other ways: weighting, resampling (ensure better representation of concerned groups), or transformation (transform data with function preserving data while hiding bias). Acting on the model. At training time: (<i>in-processing</i>) optimise accuracy and fairness together, add fairness metrics; or (<i>post-processing</i>) directly apply transformations to output, independent from model. To improve fairness of model, almost always face dilemmas: → Leveling down. More fairness for some might compromise fairness to others. → Fairness–accuracy. More fairness may often compromise accuracy. → Fairness–privacy. Access to sensitive attributes, increasing privacy may compromise fairness.	
Datasheet for dataset. Fill form with technical details/explanations about data, collection process, processing, intended usage→for devs: avoid publishing ill data; for users: ensure data cleanliness, better understand dataset. Model card. Same idea but for ML → technical info, intended use, performance, metrics, data for training/e-valuation, quantitative analyses, ethical reviews, usage recommendations.	
AI Ethics card (IDEO). Guide ethically responsible, culturally considerate design with data. Principles: data ≠ truth. don't presume AI durability; respect privacy/collective good; unintended AI consequences → design opportunities. Goal: humanize data (<i>people behind data</i>). Method: find/read documentation, raw data, materials; select variables/columns; pick few; inspect their data; focus on one variable (select people random or by specific characteristics); write conclusion in terms of ethical impact and risks.	